RAMANUJAN'S DIMENSIONAL FORCING: A UNIVERSAL FORMULA FOR SIEVE LEVELS

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ABSTRACT. We study the formula

$$\theta_k = \frac{2^k - k}{2^k} = 1 - \frac{k}{2^k},$$

proposed as the level of distribution of k-fold sieves with constitutional structure. The formula rests on a counting postulate: a system of depth k is one in which exactly k of the 2^k binary configurations are eliminated, which gives configuration density $(2^k - k)/2^k$. This is a postulate, not a consequence of the binary model: if a configuration is invalid whenever some constraint fails, there are $2^k - 1$ invalid configurations, not k. For $k = 3$ the formula gives $5/8$. We record the Eisenstein-integer arithmetic attached to the moduli $q = 3^K$ (the ramification $(3) = (1-\omega)^2$ and the cyclicity of $(\mathbb{Z}/3^K\mathbb{Z})^*$) and explain why it does not yield a level of distribution $5/8$: the summed estimate over the moduli 3^K holds for every exponent up to 1 by a trivial argument, while the per-modulus statement would go beyond what the Generalised Riemann Hypothesis gives. The value $5/8$ for cascade moduli is therefore a conjecture. Pascadi's exponent $5/8$ for general moduli with triply-well-factorable weights has a different origin, and its agreement with θ_3 is a numerical coincidence. The correspondence between θ_k and analytic levels of distribution is conjectural for every k, and for $k = 4, 5$ no object to which a prediction would apply has been specified.

Revision note (November 2026). The Counting Theorem of the first version is now a postulate (Postulate 3.4): its induction proof was wrong, and under the paper's own binary model there are $2^k - 1$ invalid configurations, not k. The "geometric proof" of $\theta_3 = 5/8$ is withdrawn: the filtration has $2K$ steps, not three; the Kloosterman bound $q^{3/8}$ is false; and the summed estimate of the former Theorem 5.7 holds for every exponent up to 1, so $5/8$ for cascade moduli is now stated as a conjecture. The claims that the formula unifies Bombieri–Vinogradov and Pascadi, that Khayyam's Triangle is its combinatorial origin, and that $\theta_5 = 27/32$ together with GPY would give infinitely many twin primes are withdrawn, as is the "empirical" column of Table 2, for which no method, code or data existed. The assignment $k = 3$ to twin primes and the Sophie Germain "non-instance" are withdrawn. The pattern-density discussion is corrected (the difference is about 9%, and the large-sieve claim is withdrawn), the "Ramanujan principle" is removed, the appendix on Watt's theorem and Condition W3 is corrected, and the Bombieri–Vinogradov and Pascadi references are fixed.

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1. Introduction

1.1. **The formula and its meaning.** The Bombieri–Vinogradov theorem [1, 15] states that for every $A > 0$ there is $B = B(A)$ such that

$$\sum_{q \le x^{1/2}(\log x)^{-B}} \max_{\gcd(a,q)=1} \left| \psi(x;q,a) - \frac{x}{\varphi(q)} \right| \ll \frac{x}{(\log x)^A},$$

where $\psi(x;q,a) = \sum_{n \le x,\ n \equiv a\,(q)} \Lambda(n)$. This is an average over moduli; in this sense primes have level of distribution $\theta_{BV} = 1/2$. Levels beyond $1/2$ are known in modified settings (well-factorable weights and a fixed residue class, or smooth moduli): Bombieri, Friedlander and Iwaniec [2] obtained $4/7$ in 1986; Zhang [17] obtained $1/2 + 1/584$ for smooth moduli, improved to $1/2 + 7/300$ by Polymath [12]; and for triply-well-factorable weights Maynard [9] obtained $3/5$, Lichtman [6] $66/107$, and Pascadi [11] $5/8$, for general moduli.

This paper studies the formula

$$\theta_k = \frac{2^k - k}{2^k}, \tag{1}$$

proposed as a universal formula for the level of distribution as a function of sieve depth k. It gives $\theta_1 = \theta_2 = 1/2$, $\theta_3 = 5/8$ and $\theta_4 = 3/4$. Formula (1) is not derived: it follows from a counting postulate (Postulate 3.4), and its relation to analytic levels of distribution is conjectural (Conjecture 6.4). The value $\theta_3 = 5/8$ agrees with Pascadi's exponent; we regard this as a numerical coincidence (Section 6.3).

1.2. **About the title.** The title is an homage to Ramanujan. The first version of this subsection attributed to him a "principle" that certain numbers force exact asymptotic formulas through their structural properties alone; no such principle appears in his work, and no result of Ramanujan is used in this paper. (His partition congruences, such as $p(5n+4) \equiv 0 \pmod 5$ [14], are congruences rather than asymptotic formulas; the asymptotic formula for the partition function is due to Hardy and Ramanujan [5].)

The first version also argued that the prime 3 forces $\theta = 5/8$: twin primes require $p \equiv 2 \pmod 3$ for $p > 3$, and this was taken to give three constitutional constraints on a configuration space with $2^3 = 8$ elements. That argument is withdrawn. For $p > 3$, twin primes satisfy the single condition $p \equiv 2 \pmod 3$; the conditions modulo 9 and 27 are implied by it, so by the independence requirement of Definition 3.2(b) the depth is 1, not 3 (Example 3.3).

1.3. **Logical status.** We state the logical status of every significant claim.

[Postulate] Postulate 3.4: a system of depth k has exactly k eliminated configurations among 2^k. This is not a theorem: in the binary model of Definition 3.1 the number of invalid configurations is $2^k - 1$ (Proposition 3.5, proved). The configuration density $(2^k - k)/2^k$ (Proposition 3.6) follows from the postulate.

[Proved] Proposition 4.1: the numbers 416/729 and 5/8 differ by about 9%, and the first is not a density of residue classes. The earlier claim that the large sieve "connects to" configuration density is withdrawn.

[Proved] Proposition 2.3: the proportion of nonzero entries in the first 3^K rows of Khayyam's Triangle mod 3 is $2^{K+1}/(3^K + 1) \to 0$. It does not match $(2^K - K)/2^K$, and the triangle is not a combinatorial origin of formula (1).

[Proved; trivial] Proposition 5.7: the summed estimate over the moduli 3^K (Theorem 5.7 of the first version) holds for every exponent up to 1, so it carries no information about 5/8.

[Conjectured] Conjecture 5.8: per-modulus equidistribution for $3^K \le x^{5/8-\varepsilon}$. This would go beyond the Generalised Riemann Hypothesis.

[Conjectured] Conjecture 6.4: θ_k equals an analytic level of distribution for general k.

[Withdrawn] The "geometric proof" of $\theta_3 = 5/8$ (Remark 5.6); the twin-prime threshold conjecture (Remark 6.5); the empirical column of Table 2; the Sophie Germain non-instance (Remark 6.7).

1.4. **Position in the programme.** This paper is the fifth in a nine-paper programme (p01–p09) on constitutional sieve theory [18, 19, 20, 22, 21, 23]. The counting formula is introduced in [18], where it is also a postulate. The level 5/8 for the moduli 3^K is a conjecture of [22]. Condition W3 of [23] is true, with a one-line proof (Appendix A), and does not bear on any level of distribution. The comparison with information theory is in [21]. The present paper discusses the formula and the distinction between two densities.

2. Khayyam's Triangle Modulo 3

Definition 2.1 (Khayyam's Triangle mod 3)**.** Khayyam's Triangle modulo 3 is the infinite triangular array with entry $\binom{n}{k} \bmod 3$ in position (n, k), $0 \le k \le n$.

Lemma 2.2 (Lucas [7])**.** *For a prime p and integers $0 \le k \le n$, write $n = \sum_i n_i p^i$ and $k = \sum_i k_i p^i$ in base p. Then*

$$\binom{n}{k} \equiv \prod_i \binom{n_i}{k_i} \pmod p.$$

In particular, $\binom{n}{k} \not\equiv 0 \pmod{p}$ *if and only if* $k_i \le n_i$ *for all* i.

Proposition 2.3 (Nonzero entries of Khayyam's Triangle mod 3)**.** *For* $K \ge 0$*, the first* 3^K *rows* ($0 \le n < 3^K$) *of Khayyam's Triangle modulo* 3 *contain exactly* 6^K *nonzero entries out of* $3^K(3^K+1)/2$*, a proportion*

$$\frac{2^{K+1}}{3^K+1}.$$

For $K = 1, 2, 3, 4, 5$ *this is* $1, 4/5, 4/7, 16/41, 16/61$*, and it tends to* 0 *as* $K \to \infty$.

Proof. Write $n = \sum_{i<K} n_i 3^i$. By Lemma 2.2 with $p = 3$, the number of k with $\binom{n}{k} \not\equiv 0 \pmod 3$ is $\prod_{i<K}(n_i+1)$. Summing over the 3^K choices of digits gives $(1+2+3)^K = 6^K$. The first 3^K rows contain $\sum_{n<3^K}(n+1) = 3^K(3^K+1)/2$ entries, and $2 \cdot 6^K/(3^K(3^K+1)) = 2^{K+1}/(3^K+1)$. □

Remark 2.4. The first version of Proposition 2.3 asserted that the Sierpiński structure partitions the configurations of a K-level cascade sieve into exactly $2^K - K$ valid and K invalid ones, matching the count of Section 3; its proof restated that count by forward reference. The assertion is withdrawn: the proportion in Proposition 2.3 tends to 0, whereas $(2^K - K)/2^K$ tends to 1, and no correspondence between nonzero entries and configurations is established. The first version also described the Sierpiński pattern and the Eisenstein filtration of Section 5 as "two faces of the same structure"; no such connection is established.

3. The Counting Postulate

3.1. Configuration space and constitutional constraints.

Definition 3.1 (Configuration space)**.** For a k-level sieve, the configuration space is $\mathcal{C}_k = \{0,1\}^k$, with $|\mathcal{C}_k| = 2^k$. Each configuration $c = (c_1, \ldots, c_k)$ records whether the constitutional constraint is satisfied ($c_i = 1$) or fails ($c_i = 0$) at level i.

Definition 3.2 (Constitutional constraint)**.** A constraint at level i is *constitutional* if:

(a) it is forced by the problem structure—not a free parameter; and
(b) it is independent of the constraints at the other levels, in the sense that it is not implied by them. (For congruence conditions to pairwise coprime moduli the Chinese Remainder Theorem guarantees this; for nested moduli such as $3, 9, 27$ it does not.)

Example 3.3 (Twin primes)**.** For a twin prime pair $(p, p+2)$ with $p > 3$: if $p \equiv 0 \pmod 3$ then $3 \mid p$; if $p \equiv 1 \pmod 3$ then $3 \mid (p+2)$. Only $p \equiv 2 \pmod 3$ permits both members to avoid divisibility by 3. This constraint satisfies Definition 3.2(a). The first version took "three nested levels" with moduli $3, 9, 27$, giving $k = 3$ and $|\mathcal{C}_3| = 8$. The conditions at the moduli 9 and 27 are not independent of the condition modulo 3: p and $p+2$ are both prime to 9 (or to 27) exactly when they are both prime to 3, so the admissible residues modulo 9 and 27 are exactly the 3 and 9 lifts of 2 mod 3. These conditions fail Definition 3.2(b), and by that criterion twin primes have depth 1. The assignment $k = 3$ to twin primes (and to the cascade moduli 3^K) is a proposal of the programme [18, 22] that is currently unsupported.

3.2. The counting postulate.

Postulate 3.4 (Nested constraint count)**.** A k-fold sieve with constitutional constraints at k nested levels is said to have *depth* k. For such a sieve we postulate that exactly k configuration classes in $\mathcal{C}_k$ are eliminated, one for each level, so that $2^k - k$ remain.

The first version stated Postulate 3.4 as a theorem, with a proof by induction on k. That proof is wrong: on passing from $k-1$ to k levels, each of the $k-1$ previously invalid configurations splits into two, so they contribute $2(k-1)$ invalid configurations, not $k-1$. The postulate is not a consequence of Definition 3.1, as the next proposition shows. It is the definition of what we mean by depth k, and the formula θ_k below is stipulated, not derived.

Proposition 3.5 (The binary model)**.** *In $\mathcal{C}_k = \{0,1\}^k$, call c* valid *if $c_i = 1$ for all i, and* invalid *otherwise (some constraint fails). Then exactly one configuration is valid and $2^k - 1$ are invalid. For each i, exactly 2^{k-1} configurations fail constraint i, and exactly k configurations fail exactly one constraint.*

Proof. The only valid configuration is $(1, \ldots, 1)$, so the other $2^k - 1$ are invalid. Fixing $c_i = 0$ leaves 2^{k-1} choices for the remaining coordinates. A configuration that fails exactly one constraint is determined by which constraint fails. □

Thus in the binary model the number of invalid configurations is $2^k - 1$ for every k (for example 7, not 3, when $k = 3$). The revised Lean file P05 proves this as `invalid_count`.

Proposition 3.6 (Configuration density)**.** *For a sieve of depth k in the sense of Postulate 3.4,*

$$\rho_k^{\text{config}} = \frac{2^k - k}{2^k}.$$

Proof. Immediate from Postulate 3.4: $2^k - k$ of the 2^k configuration classes remain. □

Table 1 displays the configuration density for $k = 1, \ldots, 6$, together with the number of invalid configurations in the binary model.

TABLE 1. Configuration density $\rho_k^{\text{config}} = (2^k - k)/2^k$ under Postulate 3.4. The last column is the number of invalid configurations in the binary model of Proposition 3.5.

k	2^k	Eliminated (postulated)	Remaining	ρ_k^{config}	$2^k - 1$ (binary)
1	2	1	1	1/2	1
2	4	2	2	1/2	3
3	8	3	5	5/8	7
4	16	4	12	3/4	15
5	32	5	27	27/32	31
6	64	6	58	29/32	63

4. Two Densities

Earlier attempts to compute θ_3 via ternary residue analysis obtained approximately 0.571 rather than $5/8 = 0.625$. The first version described this as a 12% discrepancy explained by a conflation of two densities. The difference is about 9% ($0.625/0.5706 - 1 \approx 9.5\%$, or $1 - 0.5706/0.625 \approx 8.7\%$), and the "pattern density" is the smaller of the two numbers. We record what the two numbers are.

Proposition 4.1 (Two numbers)**.** *(1) The "pattern density" of the first version is*

$$\rho_3^{\text{pattern}} = \prod_{i=1}^{3}(1 - 3^{-i}) = \frac{416}{729} \approx 0.5706.$$

This equals $|\mathrm{GL}_3(\mathbb{F}_3)|/3^9$, the probability that a random 3×3 matrix over $\mathbb{F}_3$ is invertible. It is not a proportion of residue classes modulo 27: the proportion of residues modulo 27 prime to 3 is $18/27 = 2/3$, and the proportion admissible for twin primes (p and $p+2$ both prime to 3) is $9/27 = 1/3$.

(2) Under Postulate 3.4, $\rho_3^{\text{config}} = 5/8 = 0.625$. In the binary model of Proposition 3.5, in which a configuration is valid when all its coordinates equal 1, the proportion of valid configurations is $1/2^k$ (that is, $1/8$ for $k = 3$), and $2^k - 1$ configurations are invalid.

In particular $\rho_3^{\text{pattern}} < \rho_3^{\text{config}}$.

Proof. (1) $(2/3)(8/9)(26/27) = 416/729$, and $|\mathrm{GL}_3(\mathbb{F}_3)| = (3^3-1)(3^3-3)(3^3-3^2) = 26 \cdot 24 \cdot 18 = 27 \cdot 416$, so $|\mathrm{GL}_3(\mathbb{F}_3)|/3^9 = 416/729$. There are $\varphi(27) = 18$ residues modulo 27 prime to 3, and 9 of them are $\equiv 2 \pmod 3$. (2) is Postulate 3.4 and Proposition 3.5. □

Remark 4.2. The first version asserted that the Montgomery–Vaughan large sieve [10] "connects to configuration density, not pattern density, because Dirichlet characters are indexed by constraint-satisfaction classes". This is withdrawn. The large sieve inequality sums over all primitive characters modulo q for every $q \le Q$; the characters modulo q are indexed by the dual group of $(\mathbb{Z}/q\mathbb{Z})^*$, not by binary constraint classes, and there is no "compatible" subfamily. The first version's proof also defined configuration density by "all coordinates equal to 1", which is the binary model of Proposition 3.5 and gives $1/2^k$, not $(2^k - k)/2^k$. No relation between either number and a level of distribution is established; the conjectured relation for ρ_k^{config} is Conjecture 6.4.

5. Eisenstein Integers and the Moduli 3^K

The first version of this section claimed a "geometric proof" that, for the cascade moduli $q = 3^K$, the analytic level of distribution is exactly $5/8$, via two structural facts about $\mathbb{Z}[\omega]$ each contributing half the improvement over the Weil bound. That claim is withdrawn. This section keeps the correct Eisenstein-integer arithmetic, states what is true about the summed estimate (Proposition 5.7: it holds for every exponent up to 1), and records the per-modulus statement as a conjecture (Conjecture 5.8). The steps of the withdrawn argument are listed in Remark 5.6.

5.1. The Eisenstein integers and ramification of 3.

Let $\omega = e^{2\pi i/3}$, so $\omega^2 + \omega + 1 = 0$. The Eisenstein integers are $\mathbb{Z}[\omega] = \{a + b\omega : a, b \in \mathbb{Z}\}$ with norm $\mathrm{N}(a + b\omega) = a^2 - ab + b^2$.

Lemma 5.1. $\mathrm{N}(1 - \omega) = 3$.

Proof. $\mathrm{N}(1 - \omega) = 1 - (1)(-1) + (-1)^2 = 3$. □

Proposition 5.2 (Ramification of 3)**.** *In $\mathbb{Z}[\omega]$, the prime 3 ramifies: $(3) = (1 - \omega)^2 \cdot u$ for a unit u.*

Proof. Since $1 + \omega + \omega^2 = 0$, we have $3 = -\omega^2(1 - \omega)^2$. □

A rational prime $\ell > 3$ is *inert* in $\mathbb{Z}[\omega]$ (stays prime) if $\ell \equiv 2 \pmod 3$, and *splits* as $\ell = \pi\bar{\pi}$ if $\ell \equiv 1 \pmod 3$. By Example 3.3, every twin prime $p > 3$ satisfies $p \equiv 2 \pmod 3$ (inert) while $p + 2 \equiv 1 \pmod 3$ (splits). So twin pairs with $p > 3$ are inert–split pairs; this restates $p \equiv 2 \pmod 3$ and adds no further constraint.

5.2. The $(1-\omega)$-adic filtration.

Proposition 5.3 (The $(1-\omega)$-adic filtration)**.** *The ring $\mathbb{Z}[\omega]/(3^K)$ carries the filtration*

$$\mathbb{Z}[\omega] \supset (1-\omega) \supset (1-\omega)^2 \supset \cdots \supset (1-\omega)^{2K},$$

with $(3^K) = (1-\omega)^{2K}$ and each quotient $(1-\omega)^j/(1-\omega)^{j+1} \cong \mathbb{Z}/3\mathbb{Z}$.

Proof. From Proposition 5.2, $(3) = (1-\omega)^2$, so $(3^K) = (1-\omega)^{2K}$. Each successive quotient is $\mathbb{Z}[\omega]/(1-\omega) \cong \mathbb{Z}/3\mathbb{Z}$. □

The filtration has $2K$ steps, each with quotient $\mathbb{F}_3$; nothing in it singles out three levels. (The first version called it a "three-level filtration".)

5.3. Cyclicity of $(\mathbb{Z}/3^K\mathbb{Z})^*$ and Kloosterman sums.

Lemma 5.4 (Cyclic structure)**.** $(\mathbb{Z}/3^K\mathbb{Z})^* \cong \mathbb{Z}/(2 \cdot 3^{K-1}\mathbb{Z})$ *for all $K \geq 1$.*

Proof. $\varphi(3^K) = 2 \cdot 3^{K-1}$; the unit group of $\mathbb{Z}/p^K\mathbb{Z}$ is cyclic for all odd prime powers p^K. □

Since $(\mathbb{Z}/p^K\mathbb{Z})^*$ is cyclic for every odd prime p, cyclicity does not distinguish 3 among the odd primes. The first version contrasted it with squarefree moduli, claiming that a product of unit groups "obstructs the uniform character cancellation that a single cyclic group enables". That claim is unsupported and is withdrawn: Weil's bound for Kloosterman sums holds for squarefree moduli as well, and the known levels beyond $1/2$ (Section 6.3) concern general moduli.

For integers a, b and $q \geq 1$, let $S(a,b;q) = \sum_{x \in (\mathbb{Z}/q\mathbb{Z})^*} e\big((ax + b\bar{x})/q\big)$ be the Kloosterman sum, where $\bar{x}$ is the inverse of x modulo q and $e(t) = e^{2\pi i t}$.

Proposition 5.5 (The Kloosterman bound $q^{3/8}$ fails)**.** *For every $q \geq 1$,*

$$\sum_{a \bmod q} |S(a,1;q)|^2 = q\,\varphi(q).$$

Consequently $\max_a |S(a,1;q)| \geq \varphi(q)^{1/2}$, which for $q = 3^K$ equals $(2/3)^{1/2} q^{1/2}$. In particular no bound $|S(a,b;3^K)| \ll (3^K)^{3/8+\varepsilon}$ uniform in a, b can hold.

Proof. Expanding the square and summing over a first,

$$\sum_{a \bmod q} |S(a,1;q)|^2 = \sum_{x,y \in (\mathbb{Z}/q\mathbb{Z})^*} e\Big(\frac{\bar{x} - \bar{y}}{q}\Big) \sum_{a \bmod q} e\Big(\frac{a(x-y)}{q}\Big) = q \sum_{x \in (\mathbb{Z}/q\mathbb{Z})^*} 1 = q\,\varphi(q).$$

The sum has q terms, so the largest $|S(a,1;q)|^2$ is at least $\varphi(q)$. □

Remark 5.6 (The withdrawn argument). The first version derived $\theta_3 = 5/8$ for cascade moduli from the following steps, all of which are withdrawn.

(i) A "three-level filtration" of $\mathbb{Z}[\omega]/(3^K)$. The filtration has $2K$ steps (Proposition 5.3).

(ii) A corollary stating that the filtration induces a canonical decomposition $w(d) = \alpha(d_1)\beta(d_2)\gamma(d_3)$, $d_i = 3^{j_i}$, of sieve weights on powers of 3, called triply well-factorable. The factorisation $3^j = 3^{j_1}3^{j_2}3^{j_3}$ is not unique; triple well-factorability in the sense of Maynard, Lichtman and Pascadi is the condition that for every factorisation $D = Q_1Q_2Q_3$ the weight is a Dirichlet convolution $\alpha * \beta * \gamma$ with α, β, γ supported on $[1, Q_1]$, $[1, Q_2]$, $[1, Q_3]$; and weights supported on powers of 3 do not detect primes.

(iii) A bound $E_{II} \ll x^{1/2}Q^{1/2} \cdot Q^{1/4+\varepsilon}$ for the Type II error, described as giving a "ceiling $\theta \le 2/3$". No derivation was given, and a range in which a method gives a power saving is a lower bound for the level, not a ceiling.

(iv) The Kloosterman bound $|S(a, b; 3^K)| \ll (3^K)^{3/8+\varepsilon}$, which is false (Proposition 5.5). Its proof sketch invoked Watt's theorem, which concerns the Riemann zeta function and not characters modulo q (Appendix A).

(v) The claim that (iv) "improves 2/3 to 5/8". Since $5/8 < 2/3$, this would lower the level; and even granting (iv), the stated bound $x^{1/2}Q^{1/2} \cdot Q^{3/16+\varepsilon} = x^{1/2}Q^{11/16+\varepsilon}$ gives a power saving for $Q \le x^{8/11-\delta}$, not $x^{5/8}$.

5.4. What is true about the moduli 3^K.

Proposition 5.7 (The summed estimate is trivial)**.** *For every $A > 0$,*

$$\sum_{\substack{K \ge 1 \\ 3^K \le x}} \max_{\gcd(a,3)=1} \left| \psi(x; 3^K, a) - \frac{x}{\varphi(3^K)} \right| \ll_A \frac{x}{(\log x)^A}.$$

In particular the estimate holds with the range $3^K \le x^{5/8-\varepsilon}$ of Theorem 5.7 of the first version replaced by $3^K \le x^\theta$ for any $\theta \le 1$.

Proof. Let $L = (\log x)^{A+1}$. For $3^K \le L$ the Siegel–Walfisz theorem gives $\max_a |\psi(x; 3^K, a) - x/\varphi(3^K)| \ll_A x \exp(-c\sqrt{\log x})$, and there are $O(\log\log x)$ such K. For $3^K > L$, the trivial bounds $\psi(x; q, a) \le (x/q + 1)\log x$ and $x/\varphi(3^K) = 3x/(2 \cdot 3^K)$ give $|\psi(x; 3^K, a) - x/\varphi(3^K)| \le (x/3^K + 1)\log x + 3x/(2 \cdot 3^K)$. Summing this geometric series over the $O(\log x)$ values of K with $L < 3^K \le x$ gives $\ll x \log x/L + (\log x)^2 \ll x/(\log x)^A$. □

Proposition 5.7 uses only the Siegel–Walfisz theorem and the trivial bound: no Eisenstein-integer arithmetic, Kloosterman sums or Condition W3. Theorem 5.7 of the first version is therefore true, but its exponent 5/8 plays no role, and the assertions that 5/8 is "uniquely forced" and that the filtration and cyclicity "force 5/8 and nothing else" are withdrawn. The sum involves only $O(\log x)$ moduli, so it is not a level of distribution in the usual sense. The meaningful statement is per modulus.

Conjecture 5.8 (Cascade moduli; cf. [22])**.** *For every $\varepsilon > 0$ and $A > 0$, uniformly for $q = 3^K \le x^{5/8-\varepsilon}$,*

$$\max_{\gcd(a,3)=1} \left| \psi(x; q, a) - \frac{x}{\varphi(q)} \right| \ll_{\varepsilon,A} \frac{x}{\varphi(q)(\log x)^A}.$$

For a single modulus the Generalised Riemann Hypothesis gives this only for $q \le x^{1/2-\varepsilon}$, so Conjecture 5.8 would go beyond GRH. The best known results for prime-power moduli are those of Barban–Linnik–Tshudakov (exponent 3/8), Gallagher (2/5) and Huxley (5/12); Banks and Shparlinski obtain lower bounds of the correct order for $\psi(x; q, a)$ for prime-power moduli up to $x^{0.4736}$. Montgomery's conjecture predicts the statement for every exponent below 1, so 5/8 would be a proved exponent rather than a sharp one.

6. The Formula θ_k

6.1. Statement.

Definition 6.1 (Ramanujan's Dimensional Forcing formula)**.** For a sieve of depth k in the sense of Postulate 3.4, set

$$\theta_k := \rho_k^{\text{config}} = \frac{2^k - k}{2^k}.$$

The first version stated this as a theorem which asserted, in addition, that θ_k equals the analytic level of distribution when sieve weights are k-fold factorable over cascade moduli, citing Bombieri–Vinogradov for $k = 1$ and Pascadi together with its own Theorem 5.7 for $k = 3$. Those assertions are withdrawn. The equality with an analytic level is unproved for every k and is the content of Conjecture 6.4; for $k = 3$ see Section 5, and for the comparison with known levels see Section 6.3.

6.2. The Fibonacci remark.

Proposition 6.2 (Fibonacci identity)**.** $\theta_3 = 5/8 = F_5/F_6$*, where* F_n *is the* n*-th Fibonacci number.*

Proof. $F_5 = 5$, $F_6 = 8$; $F_5/F_6 = 5/8$. □

Remark 6.3. The value $\theta_3 = 5/8 = 0.625$ lies just above $1/\varphi \approx 0.618$, where $\varphi = (1 + \sqrt{5})/2$ is the golden ratio, with $|\theta_3 - 1/\varphi| \approx 0.007$. Formula (1) does not involve Fibonacci numbers; the coincidence at $k = 3$ is noted without over-interpretation.

6.3. Comparison with known levels.

Table 2 compares formula (1) with known results.

Table 2. Formula (1) and known results.

k	Formula	Known result	Status
1	1/2	Bombieri–Vinogradov: level 1/2 for primes, on average over q [1, 15]	Numerical agreement; the 1/2 comes from the large sieve, not from a constraint count
3	5/8	Pascadi: 5/8 for general moduli with triply-well-factorable weights [11]	Numerical coincidence; for the moduli 3^K, conjectural (Conjecture 5.8)
4	3/4	—	No object specified
5	27/32	—	No object specified

The first version of Table 2 contained an "empirical" column (0.624 ± 0.002, 0.749 ± 0.003 and 0.843 ± 0.004 for $k = 3, 4, 5$), attributed to "prime distribution over $[2, 10^6]$ in cascade moduli". No statistic, method or code was given, the data described do not depend on k, and no object is specified for $k = 4, 5$. The column is withdrawn. The first version also listed Zhang (2014) as a proof of $\theta_1 = 1/2$; Zhang's level exceeds 1/2.

Levels beyond 1/2 are known in modified settings: 4/7 for well-factorable weights [2]; $1/2 + 1/584$ [17] and $1/2 + 7/300$ [12] for smooth moduli; and 3/5 [9], 66/107 [6] and 5/8 [11] for triply-well-factorable weights. Every value of θ_k has a power of 2 as its denominator, so none of 4/7, 3/5, 66/107, $1/2 + 1/584 = 293/584$ and $1/2 + 7/300 = 157/300$ is of the

form $(2^k - k)/2^k$; all of them lie strictly between $\theta_2 = 1/2$ and $\theta_3 = 5/8$. The same weight class gives 3/5, 66/107 and 5/8 through different analytic inputs, so the exponent is not determined by a count of constraints. Pascadi's 5/8 is obtained from large-sieve inequalities for exceptional Maass forms together with the Kim–Sarnak bound; it concerns general moduli and not the moduli 3^K, and its agreement with θ_3 is a numerical coincidence, not evidence for formula (1).

6.4. **Consequences and conjectures.**

Conjecture 6.4 (General forcing)**.** *For any k-fold constitutional sieve of depth k (Postulate 3.4), the analytic level of distribution equals the configuration density $\theta_k = (2^k - k)/2^k$.*

Conjecture 6.4 is not yet a precise statement: no system has been shown to have depth $k \geq 2$ in the sense of Definition 3.2, and the sequence and family of moduli whose level is meant have not been specified. For $k = 3$ the programme proposes the moduli 3^K (Conjecture 5.8); for $k \geq 4$ no candidate has been proposed.

Remark 6.5 (The twin-prime threshold conjecture is withdrawn). The first version stated a conjecture that $\theta_5 = 27/32$ holds analytically and that, combined with the Goldston–Pintz–Yıldırım theorem [4], this would imply infinitely many twin primes. The implication is false, and the conjecture is withdrawn. No level of distribution is known to imply infinitely many twin primes: a level $\theta > 1/2$ gives bounded gaps between primes [4]; even level $1 - \varepsilon$ (the Elliott–Halberstam conjecture) gives gaps at most 12 [8], and at most 6 under the generalised Elliott–Halberstam conjecture [13]; and the parity barrier prevents these sieve methods from reaching gap 2. Moreover, no object of depth 5 has been specified.

Proposition 6.6 (Limit of θ_k)**.** $\lim_{k\to\infty} \theta_k = \lim_{k\to\infty} \left(1 - \dfrac{k}{2^k}\right) = 1.$

Proof. $k/2^k \to 0$ as $k \to \infty$ by l'Hôpital's rule. □

The first version described Proposition 6.6 as consistency with the Elliott–Halberstam conjecture [3]. That conjecture concerns all moduli $q \leq x^{1-\varepsilon}$, while $\theta_k < 1$ for every k, so the limit carries no evidential weight.

6.5. **The θ_k hierarchy.**

Table 3 displays the hierarchy for $k = 1, \ldots, 5$.

Remark 6.7 (Sophie Germain pairs). The first version listed Sophie Germain pairs $(p, 2p + 1)$ as a "confirmed non-instance" at $k = 2$. For $p > 3$ they satisfy the single condition $p \equiv 2 \pmod 3$, the same as twin primes. Since $\theta_1 = \theta_2 = 1/2$, assigning them depth 1 or depth 2 predicts the same value, so the example cannot discriminate between depths and is not a falsification test. A level of distribution for Sophie Germain pairs is not a defined notion, and no such result is known. The entry is withdrawn.

Appendix A. Watt's Theorem and Condition W3

The first version of this appendix "verified" three conditions W1–W3 for an application of Watt's theorem [16] to Dirichlet L-functions modulo 3^K. Watt's theorem is a mean-value estimate in the t-aspect for $\int |\zeta(1/2 + it)|^4 |M(it)|^2 \, dt$, where M is a Dirichlet polynomial; it involves no characters modulo q and has no conditions W1–W3. The "verification" of W1 and W2 is withdrawn, as is the use of Watt's theorem for Kloosterman sums (Remark 5.6(iv)).

TABLE 3. The θ_k hierarchy. Since $\theta_1 = \theta_2 = 1/2$, the value $1/2$ cannot distinguish depth 1 from depth 2.

k	θ_k	Proposed instance	Status
1	1/2	Bombieri–Vinogradov	Level 1/2 proved (1965); not derived from a constraint count
2	1/2	None established	—
3	5/8	Twin primes; cascade moduli	Unsupported: twin primes have depth 1 (Example 3.3); 5/8 for the moduli 3^K is conjectural (Conjecture 5.8)
4	3/4	None specified	No object defined
5	27/32	None specified	No object defined

Condition W3. No $L(s,\chi)$ with χ primitive modulo 3^K has a real zero in $\sigma > 1 - c/\log 3^K$. This statement is true, with a one-line proof: $(\mathbb{Z}/3^K\mathbb{Z})^*$ is cyclic of even order, so the only real non-principal character modulo 3^K is the one induced from χ_{-3} modulo 3; an exceptional zero can only belong to a real character; so any such zero would be a zero of the single fixed L-function $L(s,\chi_{-3})$, which is positive on $(0,1]$. No CM theory is needed (the first version's appeal to a zero-free region for L-functions of CM fields is withdrawn), W3 makes no result unconditional, and nothing in it is specific to 3: $(\mathbb{Z}/\ell^K\mathbb{Z})^*$ is cyclic for every odd prime ℓ. W3 is not needed for Proposition 5.7, which uses only the Siegel–Walfisz theorem.

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The title of this paper is an homage to Srinivasa Ramanujan; no result of his is used (see Section 1.2).

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